



SCUOLA POLITECNICA E DELLE SCIENZE DI BASE

Field and Service Robotics

*Final Project*

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Academic Year 2025/2026

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# Chapter 1

## Introduction

For our final technical project, we choose to work on an underwater robot, specifically an **OpenAUV** which, unlike a conventional torpedo-shaped AUV, is highly efficient even when deployed in complex and dynamic environments. Indeed, a torpedo-shaped AUV is not a fully-actuated system and suffers from limitations regarding translational movement and in-place rotation capabilities. Conversely, the OpenAUV is a **fully-actuated** system: its vertical thrusters allow it to move up and down, while its horizontal thrusters enable translational movements within the  $xy$ -plane. Currently, due to its high deployment costs, the OpenAUV is primarily utilized for research purposes. For the sake of simplicity, the origin of the body frame coincides with the center of mass (CoM), and the **NED** (North-East-Down) convention has been adopted for its  $x_b$ - $y_b$ - $z_b$  axes.

The fact that the system is fully actuated allows us to implement a **feedback linearization control strategy**, which is further simplified by the fact that the control forces  $\mathbf{f}^b$  and torques  $\boldsymbol{\tau}^b$  are completely decoupled. However, this model-based inversion approach could lack robustness if the available dynamic model is inaccurate. To mitigate this issue, a **Momentum-based Estimator** is introduced; even in the presence of unmodeled dynamics or parametric inaccuracies, these uncertainties are lumped together, treated as external disturbances, and actively compensated for. In the presence of the observer, the actual control wrench applied to the system is computed as follows:

$$\mathbf{f}^b = \mathbf{f}_{des}^b - \hat{\mathbf{f}}_{dist}^b \quad (1.1)$$

$$\boldsymbol{\tau}^b = \boldsymbol{\tau}_{des}^b - \hat{\boldsymbol{\tau}}_{dist}^b \quad (1.2)$$

In particular, the assignment involved the implementation of the dynamic and the control system for the OpenUAV, which is implemented in the Simulink file `OpenAUV_v1dist.slx`. Subsequently, the architecture was modified to assess the robustness of the employed control system by introducing the Momentum-based Estimator as mentioned above, even in the presence of non-linear disturbances in the Simulink file `OpenAUV_v2OBS.slx`.

For the design and architecture of the OpenAUV, we considered the reference article [Sha et al. \(2025\)](#).

## Chapter 2

# Design and Development of the OpenAUV

### 2.a Design of the OpenAUV

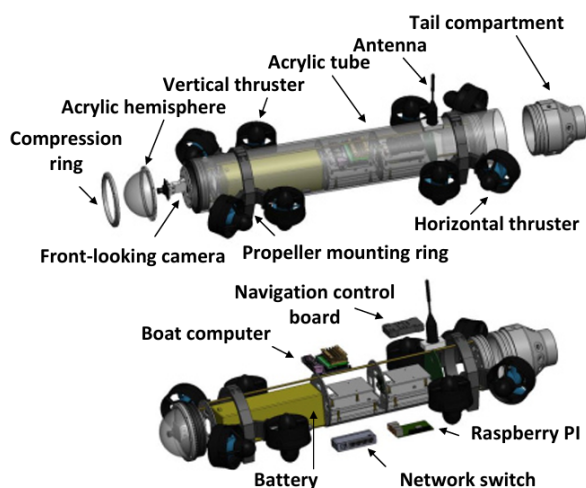


Figure 2.1: Mechanical Structure of the OpenAUV.

The OpenAUV under analysis features 8 thrusters that are directly controlled by the navigation control board. Specifically, the Vertical Thrusters are arranged perpendicular to the  $x_b$ - $y_b$  plane of the body frame to actively control heave, roll, and pitch, while the Horizontal Thrusters have a fixed inclination of  $45^\circ$  within the  $x_b$ - $y_b$  plane (**fixed orientation thrusters**) to control surge, sway, and yaw. Thanks to the combination of the aforementioned thrusters, the underwater robot is omnidirectional. The OpenAUV is equipped with several sensors, such as an antenna that enables communication with the surface, and a front-looking camera for visual servoing. Other essential sensors include an IMU to measure linear acceleration and angular velocity for estimating the vehicle's attitude, and a pressure sensor to measure the robot's depth. Additionally, sonars are integrated to detect obstacles and the seafloor, alongside a DVL (Doppler Velocity Log) to estimate the vehicle's relative velocity with respect to the seafloor. Since the system is battery-powered, energy consumption represents a critical constraint.

## 2.b Dynamic of the OpenAUV

In the implementation of the OpenAUV dynamics, in addition to the gravity effect acting along the vertical axis of the world frame, we also had to consider the buoyancy effect. The two forces act on two different points, namely the Center of Gravity (CoG)  $\mathbf{r}_g^b = [0; 0; 0.02]$  m and the Center of Buoyancy (CoB)  $\mathbf{r}_b^b = [-0.2; 0; 0.2]$  m. Since the CoB is located below the CoG, the OpenAUV features a **top-heavy** and inherently unstable **configuration**. Physically, this means that the vehicle naturally tends to rotate and capsize. This structural instability is precisely why the control architecture must be robust.

The vector  $\mathbf{g}_{rb}^b$  allows us to express the action of these two forces in a compact form as:

$$\mathbf{g}_{rb}^b = - \begin{bmatrix} \mathbf{f}_g^b + \mathbf{f}_b^b \\ \mathbf{S}(\mathbf{r}_g^b)\mathbf{f}_g^b + \mathbf{S}(\mathbf{r}_b^b)\mathbf{f}_b^b \end{bmatrix} = \begin{bmatrix} \mathbf{g}_{forces}^b \\ \mathbf{g}_{torques}^b \end{bmatrix} \in \mathbb{R}^6 \quad (2.1)$$

Following the standard NED convention, the gravitational force acts along the positive direction of the  $z$ -axis, while the buoyancy force acts in the opposite direction. The negative sign preceding the vector ensures that these hydrostatic forces and torques are correctly oriented when moved to the left-hand side of the dynamic equation.

In modifying the dynamics, we also took into account the added mass:

$$\mathbf{M}_A = \text{diag}(3.7716, 12.7006, 16.4712)$$

and the added inertia  $\mathbf{I}_{b,A} = \text{diag}(0.0545, 1.2628, 0.8198)$ , which are added to the real mass  $m = 18.5$  kg and inertia contributions of the AUV,  $\mathbf{I}_{RB} = \text{diag}(0.3432, 1.4483, 1.5047)$ , respectively. Furthermore, we had to incorporate the hydrodynamic friction forces that oppose the AUV's motion within the fluid, which are modeled through the damping matrix  $\mathbf{D}(\boldsymbol{\nu})$  as follows:

$$\mathbf{D}(\boldsymbol{\nu}) = \mathbf{D}_L + \mathbf{D}_Q(\boldsymbol{\nu})$$

with

$$\mathbf{D}_L = \begin{bmatrix} \mathbf{D}_{L,v} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{D}_{L,w} \end{bmatrix}_{6 \times 6} \quad \mathbf{D}_Q = \begin{bmatrix} \mathbf{D}_{Q,v} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{D}_{Q,w} \end{bmatrix}_{6 \times 6}$$

Specifically, Linear Damping  $\mathbf{D}_L$  dominates the system's behavior at low speeds, where the vehicle operates within a laminar flow regime. Conversely, Quadratic Damping  $\mathbf{D}_Q(\boldsymbol{\nu})$  becomes predominant when the vehicle moves at higher velocities, causing the fluid-structure interaction to transition into a turbulent flow regime.

The added mass  $\mathbf{M}_A$ , representing the volume of displaced water that opposes the body's motion, must be added to the real mass  $\mathbf{M}_{RB}$  of the underwater vehicle. The reference values for the added mass  $\mathbf{M}_A$ , the linear damping matrix  $\mathbf{D}_L$ , and the quadratic damping matrix  $\mathbf{D}_Q$  were taken from the article [Sha et al. \(2025\)](#), which analyzes the design and control of an OpenAUV. The dynamic model that we took into consideration in the code implementation is the following:

$$\mathbf{M}_V \begin{bmatrix} \dot{\mathbf{v}}_b^b \\ \dot{\boldsymbol{\omega}}_b^b \end{bmatrix} + \left( \mathbf{C}_V(\mathbf{v}_b^b, \boldsymbol{\omega}_b^b) + \mathbf{D}_{RB}(\mathbf{v}_b^b, \boldsymbol{\omega}_b^b) \right) \begin{bmatrix} \mathbf{v}_b^b \\ \boldsymbol{\omega}_b^b \end{bmatrix} + \mathbf{g}_{rb}^b = \begin{bmatrix} \mathbf{f}^b \\ \boldsymbol{\tau}^b \end{bmatrix}$$

with

$$M_V = \begin{bmatrix} M_{mass} & -mS(r_g^b) \\ mS(r_g^b) & I_b \end{bmatrix} \quad C_V(v_b^b, \omega_b^b) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -S(M_{mass}v_b^b) \\ -S(M_{mass}v_b^b) & -S(I_b\omega_b^b) \end{bmatrix}$$

$$D_{RB}(v_b^b, \omega_b^b) = \begin{bmatrix} D_v & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & D_w \end{bmatrix}$$

Specifically  $M_{mass} = mI_3 + M_A$ ,  $I_b = I_{RB} + I_{b,A}$ ,  $D_v = D_{L,v} + D_{Q,v}\|v_b^b\|$  and  $D_w = D_{L,w} + D_{Q,w}\|\omega_b^b\|$  are written in the body frame.

## 2.c Thruster Configuration Matrix Derivation

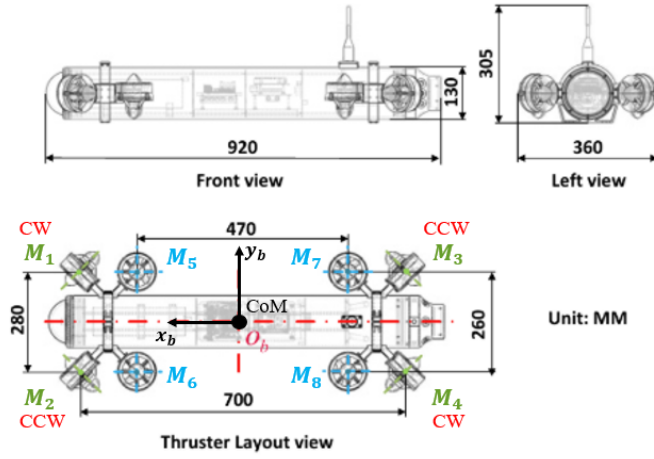


Figure 2.2: Physical Dimensions of the OpenAUV.

### 2.c.1 Analytical Derivation of the Control Allocation Matrix

To derive the **Thruster Configuration Matrix**  $B$ , we based our structural dimensions on the official technical drawings provided in the *Mechanical Structure* section of the OpenAUV repository available at <https://github.com/schahzy/OpenAUV>.

The allocation matrix  $B \in \mathbb{R}^{6 \times 8}$  maps the individual thruster forces  $\mathbf{u} = [u_1, u_2, \dots, u_8]^T$  into the total virtual control wrench vector  $\mathbf{w}^b = [f_x, f_y, f_z, \tau_x, \tau_y, \tau_z]^T$  acting on the vehicle. Mathematically, this mapping is expressed through the linear combination:

$$\mathbf{w}^b = B\mathbf{u} = \sum_{i=1}^8 \begin{bmatrix} \mathbf{d}_i \\ \mathbf{r}_i \times \mathbf{d}_i \end{bmatrix} u_i \quad (2.2)$$

The primary step in this derivation requires establishing the position vectors  $\mathbf{r}_i$ , which represent the physical distance from the center of each thruster  $M_i$  to the origin of the body-fixed frame  $O_b$ , defined as:

$$\mathbf{r}_i = [x_i, y_i, z_i]^T \in \mathbb{R}^3, \quad \forall i \in \{1, \dots, 8\} \quad (2.3)$$

in accordance with the reference convention illustrated in Figure 2.2. Notably, the  $z$ -coordinate

for all position vectors is assumed to be identically zero ( $z_i = 0$ ), under the assumption that the center of geometric thrust for all engines lies coplanar with the vehicle's CoM.

Subsequently, we defined the unit thrust vectors  $\mathbf{d}_i$ , which map the directional orientation of each actuator's thrust output:

$$\mathbf{d}_i = [d_{x,i}, d_{y,i}, d_{z,i}]^T \in \mathbb{R}^3, \quad \forall i \in \{1, \dots, 8\} \quad (2.4)$$

Regarding the horizontal thrusters ( $M_1$  to  $M_4$ ), their profiles are fixed at a  $45^\circ$  canting angle within the  $x_b$ - $y_b$  plane. Conversely, the vertical thrusters ( $M_5$  to  $M_8$ ) are aligned parallel to the vertical  $z_b$  axis. By computing the vector cross-product  $\mathbf{r}_i \times \mathbf{d}_i$  for each actuator to determine the induced moment leverage arms, the full Control Allocation Matrix  $\mathbf{B} \in \mathbb{R}^{6 \times 8}$  is explicitly structured as follows:

$$\mathbf{B} = \begin{bmatrix} 0.7071 & 0.7071 & -0.7071 & -0.7071 & 0 & 0 & 0 & 0 \\ 0.7071 & -0.7071 & 0.7071 & -0.7071 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & -1.0000 & -1.0000 & 1.0000 \\ 0 & 0 & 0 & 0 & 0.1300 & 0.1300 & -0.1300 & -0.1300 \\ 0 & 0 & 0 & 0 & -0.2350 & 0.2350 & -0.2350 & 0.2350 \\ 0.1485 & -0.1485 & -0.1485 & 0.1485 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (2.5)$$

In the fixed 8-thruster structural layout of the OpenAUV prototype, the total torque about the Yaw axis is:

$$\tau_z = \underbrace{0.1485 \cdot u_1 - 0.1485 \cdot u_2 - 0.1485 \cdot u_3 + 0.1485 \cdot u_4}_{\text{Horizontal Thrusters}} + \underbrace{c_m \cdot \sum_{j=1}^8 u_j}_{\text{Parasitic reactive torques}}$$

Due to the symmetrical distribution of the 4 vertical thrusters ( $M_5$  to  $M_8$ ), which are installed in alternating clockwise (CW) and counterclockwise (CCW) pairs, their parasitic reactive torques algebraically cancel each other out along the Roll ( $\tau_x$ ) and Pitch ( $\tau_y$ ) axes.

Thus, regarding the Yaw axis,  $\tau_z$  is dominated by the 4 horizontal thrusters ( $M_1$  to  $M_4$ ).

The relationship between the control inputs  $\mathbf{f}^b$  and  $\boldsymbol{\tau}^b$ , computed by the navigation board, and the physical actuators must be accurately modeled. The thrust  $u_i$  follows a quadratic hydrodynamic law with respect to the propeller's angular velocity  $\Omega_{des,i}$ , formulated as:

$$u_i = c_f \cdot \Omega_{des,i} |\Omega_{des,i}| \quad (2.6)$$

where  $c_f$  represents the *lumped thrust coefficient*. In real-world underwater applications, such as the *BlueRobotics T200* thrusters integrated into the OpenAUV, for the *thrust asymmetry* of the propellers:

- $c_{f,forward} = 4.7 \times 10^{-5} \text{ N} \cdot \text{s}^2/\text{rad}^2$ : This nominal coefficient is applied when the propeller spins in its primary direction, generating positive thrust ( $u_i \geq 0$ ).
- $c_{f,backward} = 2.9 \times 10^{-5} \text{ N} \cdot \text{s}^2/\text{rad}^2$ : This reduced coefficient is utilized when the motor

reverses its rotation to brake or back up, producing negative thrust ( $u_i < 0$ ).

Consequently, to map the force demands  $u_i$  from the control allocation matrix into explicit angular speed commands  $\Omega_{des,i}$  routed to the Electronic Speed Controllers (ESCs), the inversion mapping is implemented inside the simulation architecture as follows:

$$\Omega_{des,i} = \begin{cases} \sqrt{\frac{u_i}{c_{f,forward}}}, & \text{if } u_i \geq 0 \\ -\sqrt{\frac{|u_i|}{c_{f,backward}}}, & \text{if } u_i < 0 \end{cases} \quad (2.7)$$

Accounting for this asymmetry is crucial for controller synthesis. It ensures that the vehicle's braking maneuvers and reverse translations require higher angular velocities than forward operations, mirroring the exact physical limitations encountered by the physical vehicle in water.

## 2.d Feedback Linearization Control

In the OpenAUV control the following feedback linearization was implemented:

$$\mathbf{f}^b = \mathbf{M}_{mass} \mathbf{a}_{PID}^b - m \mathbf{S}(\mathbf{r}_g^b) \mathbf{a}_{PID}^b + \mathbf{D}_v(\mathbf{v}_b^b) + \mathbf{g}_{forces}^b + \mathbf{S}(\boldsymbol{\omega}_b^b) \mathbf{M}_{mass} \mathbf{v}_b^b - \hat{\mathbf{f}}_{dist}^b \quad (2.8)$$

with

$$\mathbf{a}_{PID}^b = \mathbf{R}_b^T \mathbf{a}_{PID}$$

and

$$\mathbf{a}_{PID} = \dot{\mathbf{v}}_{des} - \mathbf{K}_p \mathbf{e}_p - \mathbf{K}_d \dot{\mathbf{e}}_p - \mathbf{K}_i \int \mathbf{e}_p dt$$

$$\boldsymbol{\tau}^b = \mathbf{I}_b \mathbf{a}_{PID}^b + m \mathbf{S}(\mathbf{r}_g^b) \mathbf{a}_{PID}^b + \mathbf{D}_w(\boldsymbol{\omega}_b^b) + \mathbf{g}_{torques}^b + \mathbf{S}(\boldsymbol{\omega}_b^b)(\mathbf{I}_b \boldsymbol{\omega}_b^b) + \mathbf{S}(\mathbf{v}_b^b) \mathbf{M}_{mass} \mathbf{v}_b^b - \hat{\boldsymbol{\tau}}_{dist}^b \quad (2.9)$$

with

$$\mathbf{a}_{PID}^b = -\mathbf{K}_R \mathbf{e}_R - \mathbf{K}_\omega \mathbf{e}_\omega - \mathbf{K}_{iR} \int \mathbf{e}_R dt - \mathbf{S}(\boldsymbol{\omega}_b^b) \boldsymbol{\omega}_{des}^b + \dot{\boldsymbol{\omega}}_{des}^b$$

where:

- $\hat{\mathbf{f}}_{dist}^b$  is the force computed by the Momentum-based Estimator;
- $\hat{\boldsymbol{\tau}}_{dist}^b$  is the torque computed by the Momentum-based Estimator;
- $-m \mathbf{S}(\mathbf{r}_g^b) \mathbf{a}_{PID}^b$  and  $m \mathbf{S}(\mathbf{r}_g^b) \mathbf{a}_{PID}^b$  represent the gyroscopic and inertial coupling forces/torques induced by the CoG offset;
- $\mathbf{S}(\boldsymbol{\omega}_b^b) \mathbf{M}_{mass} \mathbf{v}_b^b$  represents the Coriolis and centripetal forces accounting for both rigid-body and hydrodynamic added mass;
- $\mathbf{S}(\mathbf{v}_b^b) \mathbf{M}_{mass} \mathbf{v}_b^b$  is the hydrodynamic moment, capturing the cross-coupling torques induced by the linear velocity and total mass matrix.



## Chapter 3

# OpenAUV under the Effect of Disturbances

The underwater robot was initially assigned a trajectory defined by the desired coordinates  $x_f = 1$  m,  $y_f = 1$  m,  $z_f = -4$  m, starting from the origin at a depth of  $-1$  m, and final orientations  $\phi_f = \pi/4$  rad,  $\theta_f = \pi/4$  rad,  $\psi_f = \pi$  rad, starting from  $\mathbf{R}_b(t_i) = \mathbf{I}_3$ . The vehicle successfully reaches the target state after approximately 22 seconds, entering a stable hovering phase for the remainder of the simulation.

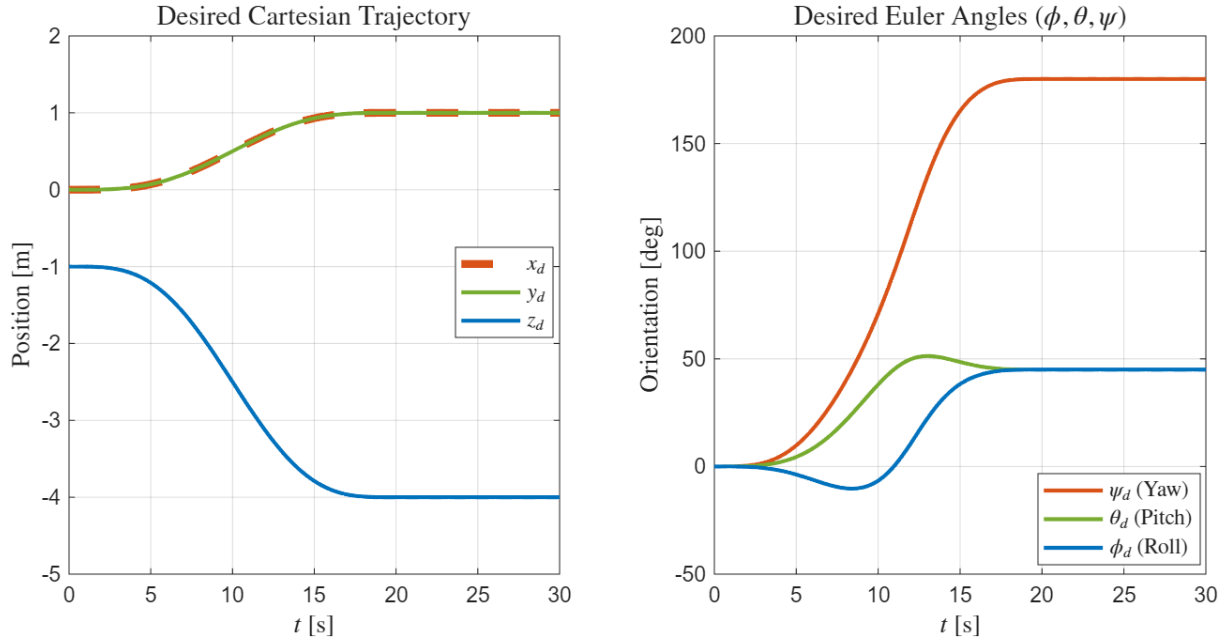


Figure 3.1: Time history of the Desired Position and Orientation of the OpenAUV.

The behavior analysis of the OpenAUV was conducted by considering the following **non-linear disturbances**:

- the **drag action**  $\mathbf{D}_Q \text{diag}(|\boldsymbol{\nu}|)\boldsymbol{\nu}$ , where  $\boldsymbol{\nu} \in \mathbb{R}^6$  is the relative twist of the CoM with respect to that of the current, expressed in the body frame. This matrix accounts for the hydrodynamic drag forces and torques acting on the system, representing the fact that the vehicle operates within a **turbulent flow regime** where the hydrodynamic resistance

scales quadratically with velocity. Specifically, the damping matrix used in the model is defined as  $\mathbf{D}_Q = \text{diag}(0.15, 0.15, 0.25, 0.02, 0.02, 0.04)$ ;

- a constant **current action**  $\boldsymbol{\nu}_c = [3, 0, 0]^T$  m/s, represented in the world frame.
- a force and a moment with a **sinusoidal trend**, having an amplitude of 8 N for the force and 2 Nm for the moment. Both sinusoidal waves have an angular frequency of 0.5 rad/s. This disturbance can originate from **time-varying currents** or **tidal** effects;
- an amplitude **step variation** between 0.5 and 1.5, starting 4 seconds after the beginning of the simulation. This step function can be used to model a time-varying ocean current.

### 3.a Behavior Analysis of the OpenAUV without the Momentum-based Estimator

#### 3.a.1 PD Control

Under a purely PD control scheme, the constant force disturbance introduced by the ocean current cannot be fully compensated. Due to hydrodynamic cross-coupling, specifically the added mass effects captured within the Coriolis matrix  $\mathbf{C}_V$ , this linear disturbance propagates into the rotational dynamics, generating constant steady-state torques. Consequently, both position and attitude errors converge to constant, non-zero values (Figure 3.2). To eliminate these steady-state offsets across the affected channels, an integral action must be integrated alongside the proportional and derivative terms.

The analysis of the OpenAUV's behavior will be primarily based on the study of the position and orientation errors, which are computed as follows:

$$\mathbf{e}_p = \mathbf{p}_b - \mathbf{p}_{b,d} \quad (3.1)$$

$$\mathbf{e}_R = \frac{1}{2} \left( \mathbf{R}_{b,d}^\top \mathbf{R}_b - \mathbf{R}_b^\top \mathbf{R}_{b,d} \right)^\vee \quad (3.2)$$

It is important to note that the individual components of the rotation matrix error  $\mathbf{e}_R$  do not directly map to the standard Euler angle errors (roll, pitch, and yaw).

The  $(\cdot)^\vee$  symbol denotes the *Vee operator*, which acts as the inverse mapping of the skew-symmetric operator  $\mathbf{S}(\cdot)$ .

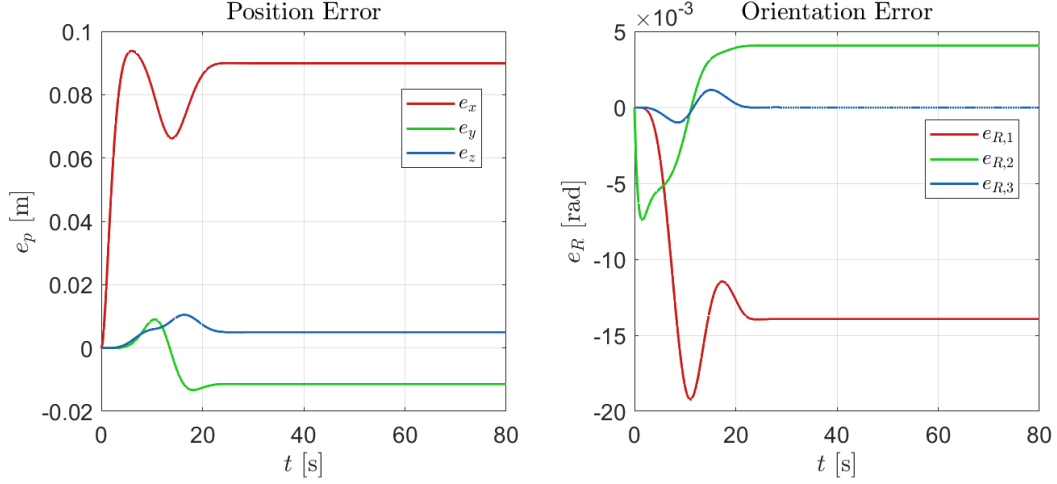


Figure 3.2: Behavior of the OpenAUV with the **constant current** and PD action.

Furthermore, it can be observed that if the current action is removed (Figure 3.3), the PD controller is fully capable of driving both the position and attitude errors to zero. This occurs because the drag force is velocity-dependent; as the AUV approaches the hovering state, the velocity vanishes, causing the hydrodynamic drag to disappear. Consequently, in the absence of external current disturbances, the steady-state error is eliminated even without integral action, as the system does not need to overcome any constant external bias.

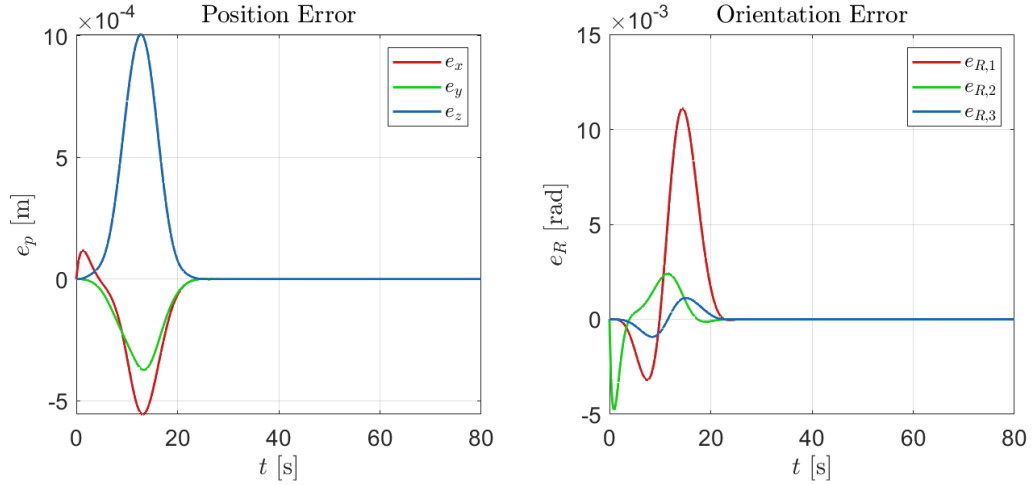


Figure 3.3: Behavior of the OpenAUV without constant current with PD action.

The matrices utilized for the PD control are defined as follows:

$$\mathbf{K}_p = \omega_{n,\text{pos}}^2 \mathbf{I}_3 \quad (3.3)$$

$$\mathbf{K}_d = 2\zeta_{\text{pos}}\omega_{n,\text{pos}}\mathbf{I}_3 \quad (3.4)$$

where  $\omega_{n,\text{pos}} = 0.8 \text{ rad/s}$  and  $\zeta_{\text{pos}} = 0.8$ .

Regarding the attitude control, the following parameters were selected for the gain matrices:

$$\mathbf{K}_R = \omega_{n,\text{att}}^2 \mathbf{I}_3 \quad (3.5)$$

$$\mathbf{K}_\omega = 2\zeta_{\text{att}}\omega_{n,\text{att}}\mathbf{I}_3 \quad (3.6)$$

with  $\omega_{n,\text{att}} = 1.2 \text{ rad/s}$  and  $\zeta_{\text{att}} = \zeta_{\text{pos}}$ .

### 3.a.2 PID Control

As previously stated, in the presence of a constant current action, an integral component must be added to the control system to eliminate the steady-state position error.

Specifically, for the six-degree-of-freedom system, the control wrench is partitioned into forces  $\mathbf{f}^b(t)$  and torques  $\boldsymbol{\tau}^b(t)$ , whose expressions are given in Equations 2.8 and 2.9, respectively. Note that, since the baseline configuration without the Momentum-based Estimator is initially considered, the estimated disturbance terms  $\hat{\mathbf{f}}_{\text{dist}}^b$  and  $\hat{\boldsymbol{\tau}}_{\text{dist}}^b$  are neglected.

For the selection of the integral gains  $\mathbf{K}_i$  and  $\mathbf{K}_{iR}$ , we initially started with a value of  $0.1\mathbf{I}_3$  (Figure 3.4). During the simulations, parameter tuning was performed until a value of  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.7\mathbf{I}_3$  provided a satisfactory steady-state response, as well as adequate convergence times (Figure 3.5). It is worth noting that further increasing the integral gain  $\mathbf{K}_i$  leads to an increase in position error oscillations; this is due to the amplified phase lag and reduced damping characteristic of the closed-loop system in a fluid environment, which causes the controller to over-correct the vehicle's trajectory.

To implement the discrete control behavior for the integral action, discrete-time integrators were introduced using the **Discrete-Time Integrator** blocks with a sampling time step of 1 ms.

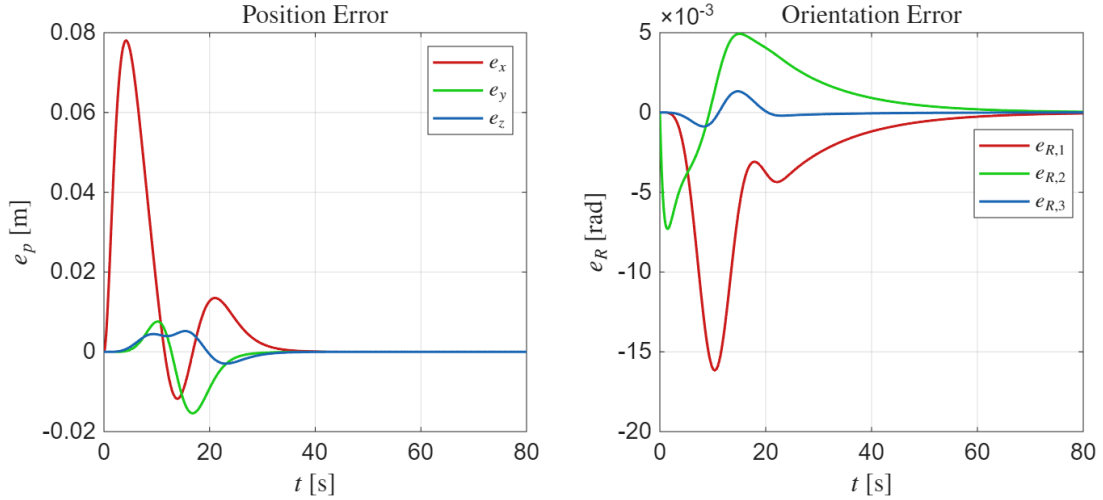


Figure 3.4: Behavior of the OpenAUV with the **current** and PID action,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

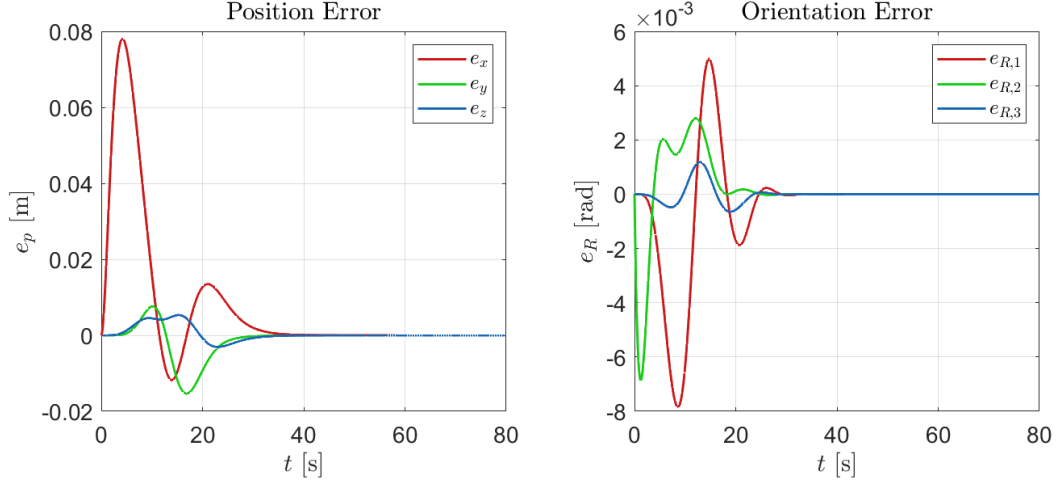


Figure 3.5: Behavior of the OpenAUV with the **current** and PID action,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.7\mathbf{I}_3$ .

In the following simulations, in addition to the constant current  $\nu_c$ , we introduced an environmental disturbance acting on both forces and moments with a **sinusoidal profile**. Regarding the integral gains  $\mathbf{K}_i$  and  $\mathbf{K}_{iR}$ , the original parameters were restored to  $0.1\mathbf{I}_3$ . It can be observed that both the position and orientation errors oscillate around zero without ever converging to a null value. This behavior occurs because the integral action guarantees error convergence to zero only for constant disturbances, and is inherently unable to completely reject time-varying perturbations.

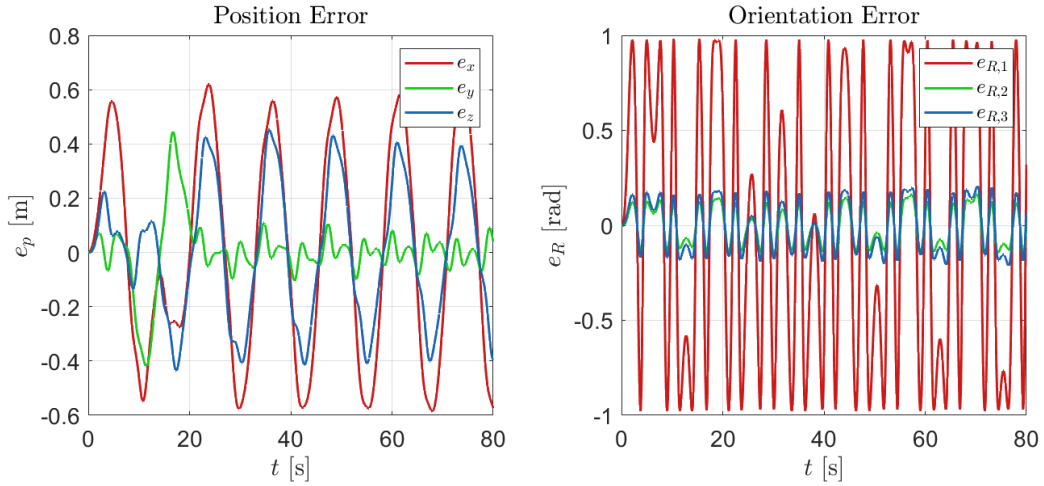


Figure 3.6: Behavior of the OpenAUV with the **current**, **sinusoidal** force and moment and PID action,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

We observe that the harmonics exhibit a non-linear behavior due to several key factors:

- The **actuator delay**, which is modeled and implemented via the **Unit Delay** block;
- The impact of the **sinusoidal waves** acting on forces and moments, which continuously affects and alters both the vehicle's position and body orientation;
- The quadratic damping matrix  $\mathbf{D}_Q$  introduces hydrodynamic drag forces and torques that vary proportionally with each velocity component multiplied by its own magnitude;

- The **frequency mismatch** between the external disturbance (0.5 rad/s) and the controlled AUV natural frequencies (0.8 rad/s and 1.2 rad/s), which triggers a physical phenomenon known as *beating*, causing the amplitude of the resulting waves to periodically increase and decrease over time.

Still assuming a constant marine current as the baseline environmental disturbance, we have now introduced a **Step** block perturbation acting simultaneously on both the linear forces and angular moments. In both cases, the step signal starts at an initial value of 0 and transitions to a final value of 0.5 N at approximately  $t = 4$  seconds. Throughout this simulation, the integral gain parameters were kept unchanged at  $\mathbf{K}_i = \mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

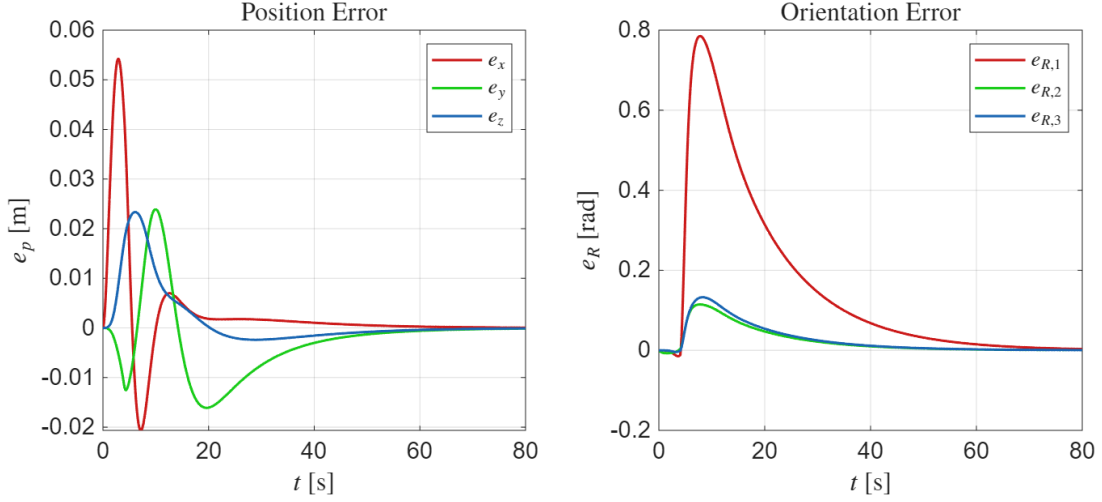


Figure 3.7: Behavior of the OpenAUV with the **current**, 0.5 **Step** force and moment and PID action,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

We observed that increasing the final amplitude of the Step input to 1.5 while keeping the integral action with the same values as before, causes the system to become unstable.

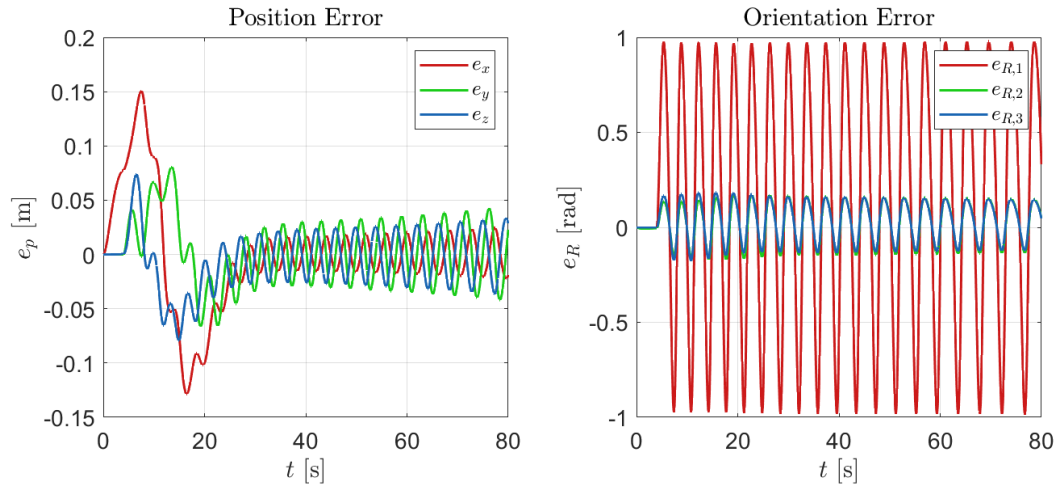


Figure 3.8: Behavior of the OpenAUV with the **current**, 1.5 **Step** force and moment and PID action,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

Consequently, if the step amplitude is increased, the gains of the  $\mathbf{K}_i$  and  $\mathbf{K}_{iR}$  matrices must be appropriately recalibrated case by case.

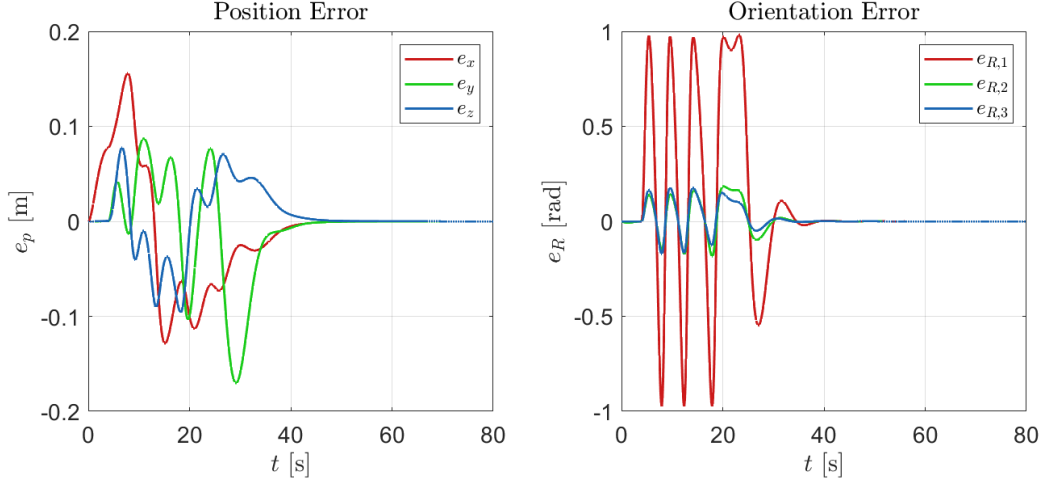


Figure 3.9: Behavior of the OpenAUV with the **current**, 1.5 **Step** force and moment and PID action,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.7\mathbf{I}_3$ .

The simulations were obtained using the Simulink file `OpenAUV_v1dist.slx` as a reference.

### 3.b Behavior Analysis of the OpenAUV with the Momentum-based Estimator

The following simulations were conducted using the Simulink model `OpenAUV_v20BS.slx`, updated with the integration of a second-order ( $r = 2$ ) Momentum-based Estimator. By keeping the integral action gains unchanged, a comparison with the previous results shown in Figure 3.5 reveals a degradation in the position and orientation tracking performance. Specifically, the position error worsens as the amplitude of the oscillations increases. Similarly, the orientation error exhibits both larger oscillatory amplitudes and significantly slower convergence times, reaching up to 60 seconds. To mitigate the convergence delay in orientation, the integral gain matrix  $\mathbf{K}_{iR}$  was selectively adjusted. As shown in Figure 3.10, the delay primarily affects the  $e_{R,1}$  component; consequently, the integral gain for this specific component was increased from 0.7 to 3. Additionally, the integral action for the remaining two orientation components was moderately increased from 0.7 to 1. Finally, it was observed that any further attempt to simultaneously increase the integral action for position tracking pushed the system to the very brink of stability.

Since the estimator must be fast,  $\lambda$  needs to be chosen sufficiently high. However, as proven during the simulations, if it was chosen too high, the system went out-of-band, causing the oscillations to become too dense. Ultimately, the selected value is  $\lambda_i = -60$  for  $i = 1, \dots, 6$ .

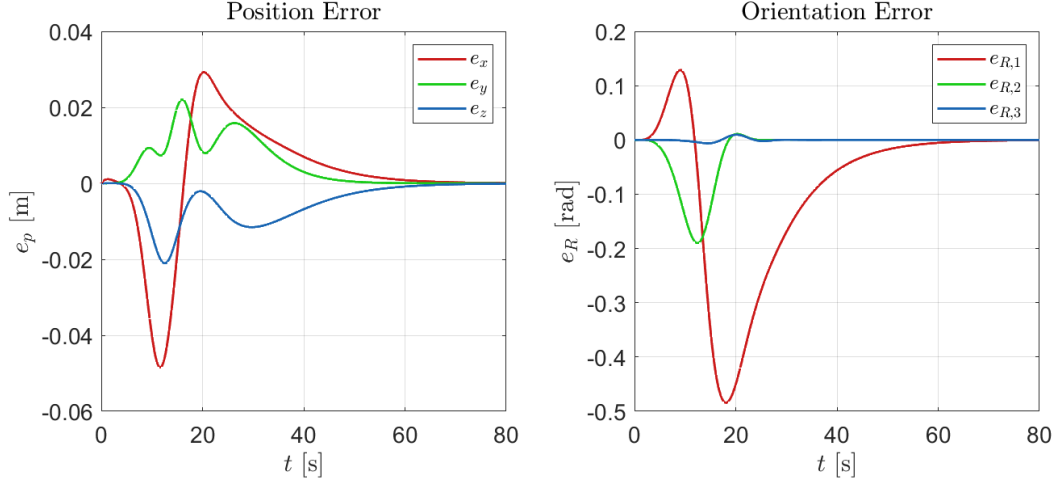


Figure 3.10: Behavior of the OpenAUV with **current** and PID action and with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.7\mathbf{I}_3$ .

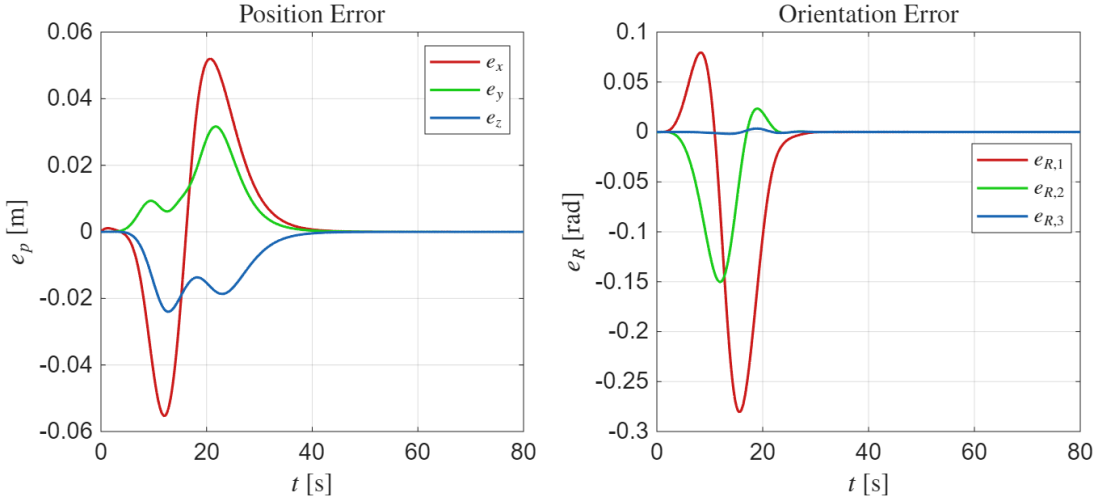


Figure 3.11: Behavior of the OpenAUV with **current** and PID action and with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ .

We now analyze the behavior of the system in the presence of both a constant marine current and sinusoidal input forces and moments acting as environmental disturbances. The amplitude and frequency of the sinusoidal waves remain unchanged compared to the previous test case (Figure 3.6). To establish a direct comparison with the scenario without the estimator, the gain matrices were kept at  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$  (Figure 3.12). Significant improvements can be observed in the oscillations, with a noticeable reduction in amplitude affecting both the position and orientation errors.



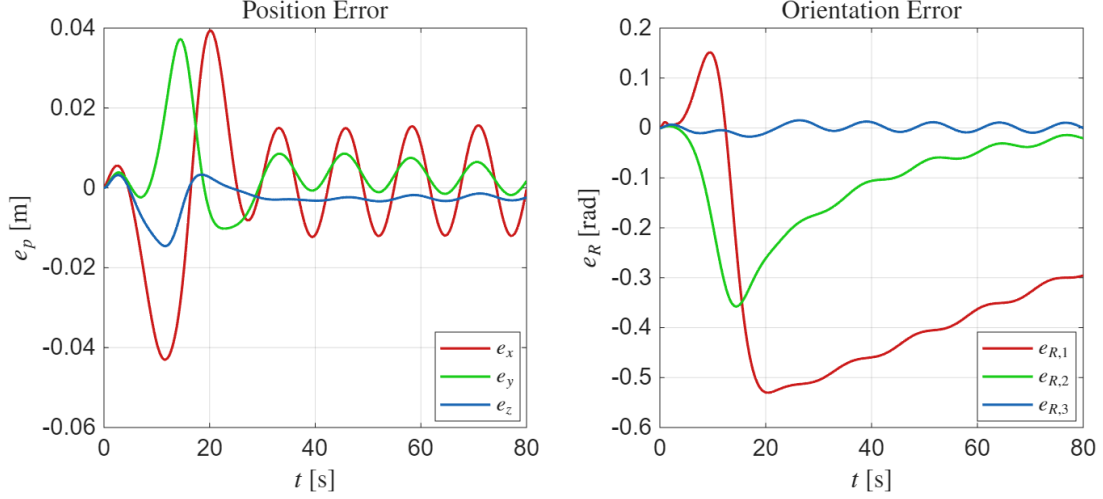


Figure 3.12: Behavior of the OpenAUV with **current**, **sinusoidal** force and moment, PID action and with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

It can be observed that, in the aforementioned case, the orientation error begins to oscillate around zero after approximately 120 seconds. This slow convergence rate can be improved by increasing the values of the gain matrix  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ .

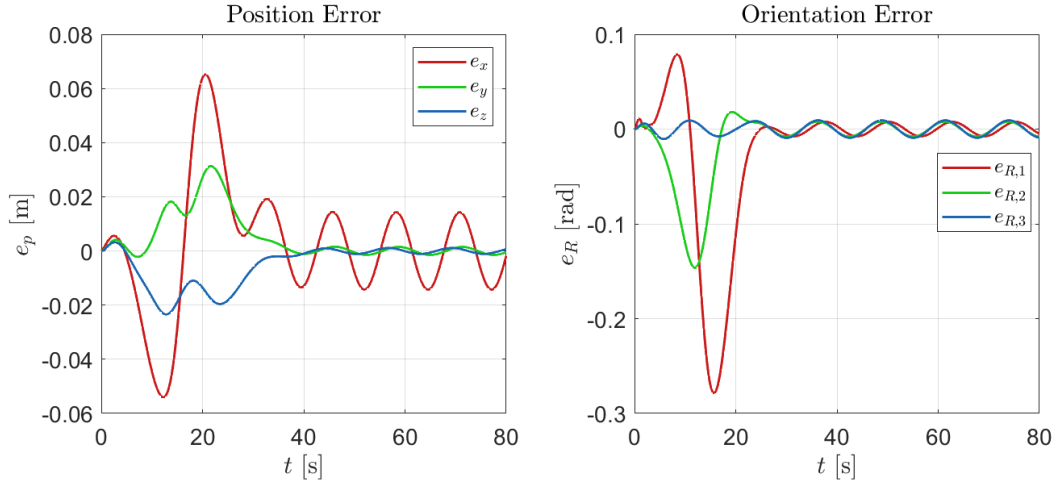


Figure 3.13: Behavior of the OpenAUV with **current**, **sinusoidal** force and moment, PID action and with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ .

We now analyze the system behavior when subjected to environmental disturbances consisting of both a constant marine current and a step input with an amplitude of 0.5 starting at  $t = 4$  seconds (Figure 3.14). Compared to the scenario without the estimator (Figure 3.7), it can be observed that the amplitude of the oscillations is reduced for the orientation error. Furthermore, a distinct transient behavior is noticeable, which is directly caused by the onset of the step perturbation at 4 seconds. Specifically, the introduction of the step input alters the acting force, thereby temporarily destabilizing both the position and orientation errors; nonetheless, they eventually converge to zero after a significant period of time.

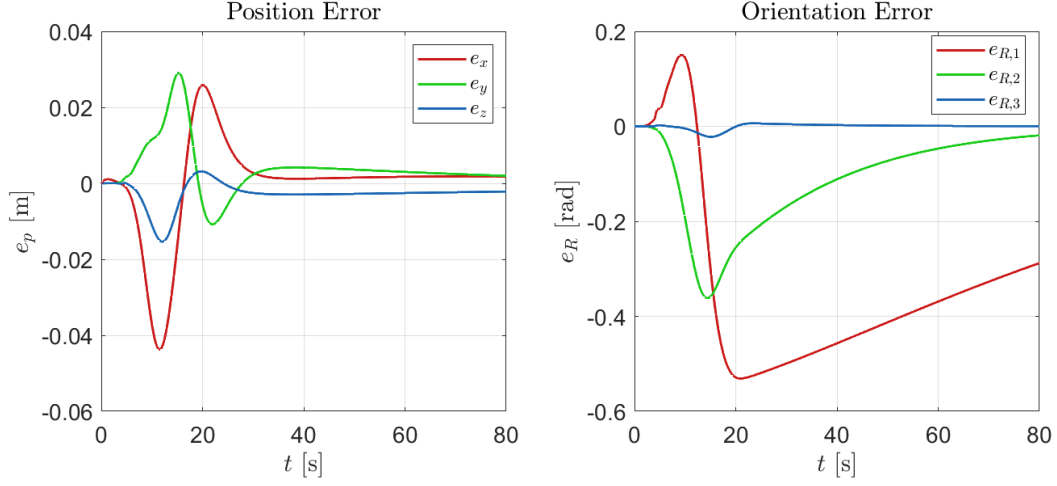


Figure 3.14: Behavior of the OpenAUV with the **current**, 0.5 **Step** force and moment and PID action with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

At the time instant  $t = 4$  seconds, the estimated force also undergoes a change in its profile, which justifies the observed variation in the orientation error behavior.

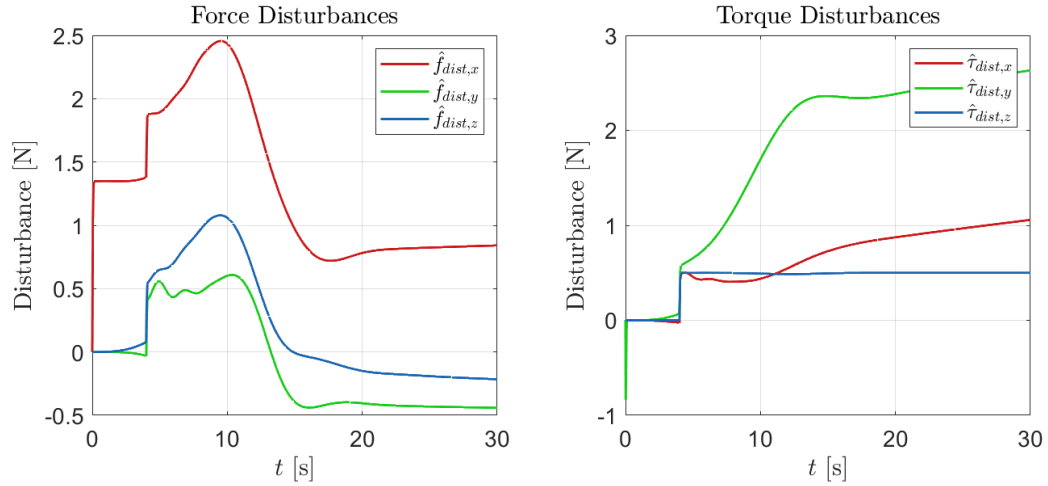


Figure 3.15: Time history of  $\hat{w}_{dist}$  of the OpenAUV with the **current**, 0.5 **Step** force and moment and PID action with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

Since the orientation error exhibits a slow convergence rate in this case as well, we modified the integral gain matrix  $\mathbf{K}_{iR}$ . This adjustment yielded good performance in terms of both oscillation amplitudes and the time required for the system to drive the error to zero.

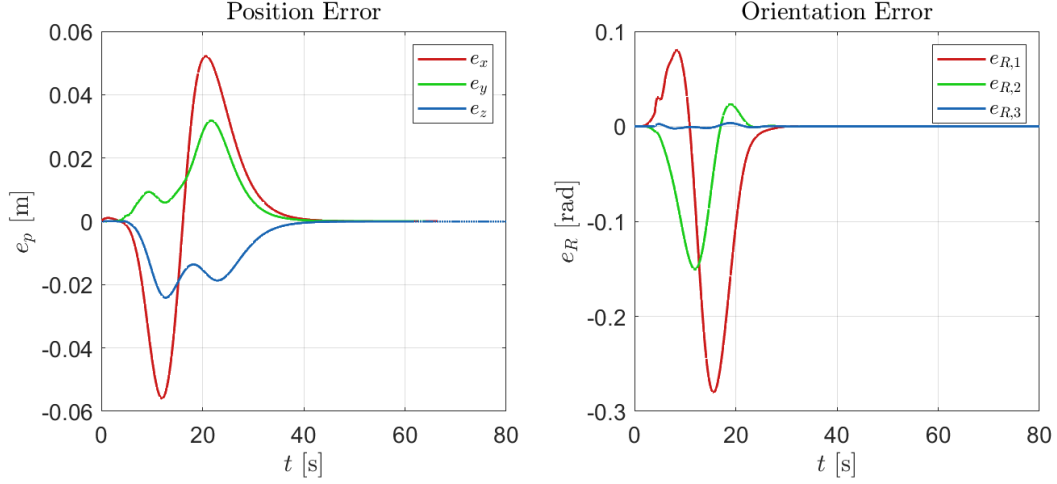


Figure 3.16: Behavior of the OpenAUV with the **current**, 0.5 **Step** force and moment and PID action with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ .

By increasing the step amplitude from 0.5 to 1.5 while keeping the onset time unchanged at 4 seconds, we can observe major improvements in the oscillations compared to the case without the estimator (Figure 3.8). However, similar to the previous scenario (Figure 3.14), the convergence time for the orientation error remains very long (Figure 3.17).

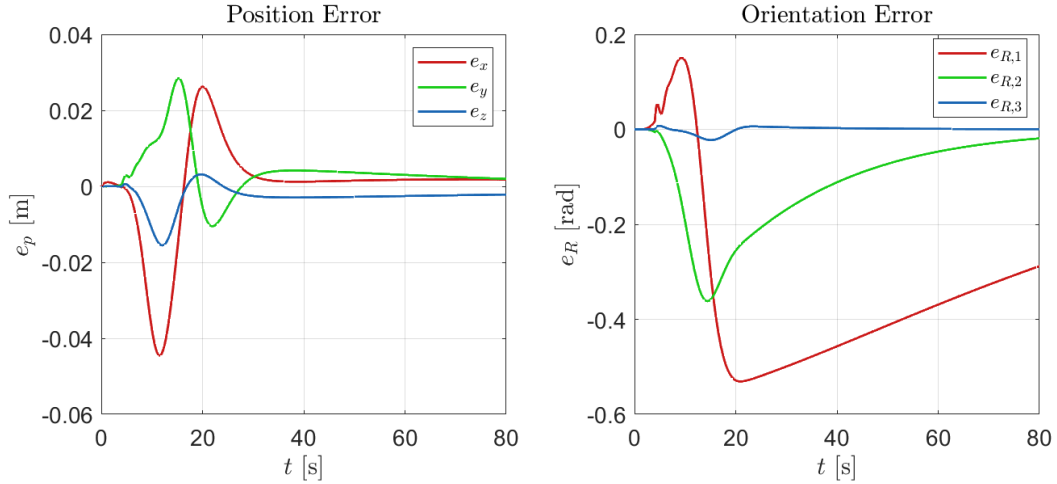


Figure 3.17: Behavior of the OpenAUV with the **current**, 1.5 **Step** force and moment and PID action with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = 0.1\mathbf{I}_3$ .

We therefore increased  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ . However, by pushing the integral gain values too far, the system exhibited persistent steady-state oscillations affecting both  $\mathbf{e}_R$  and  $\mathbf{e}_p$ .

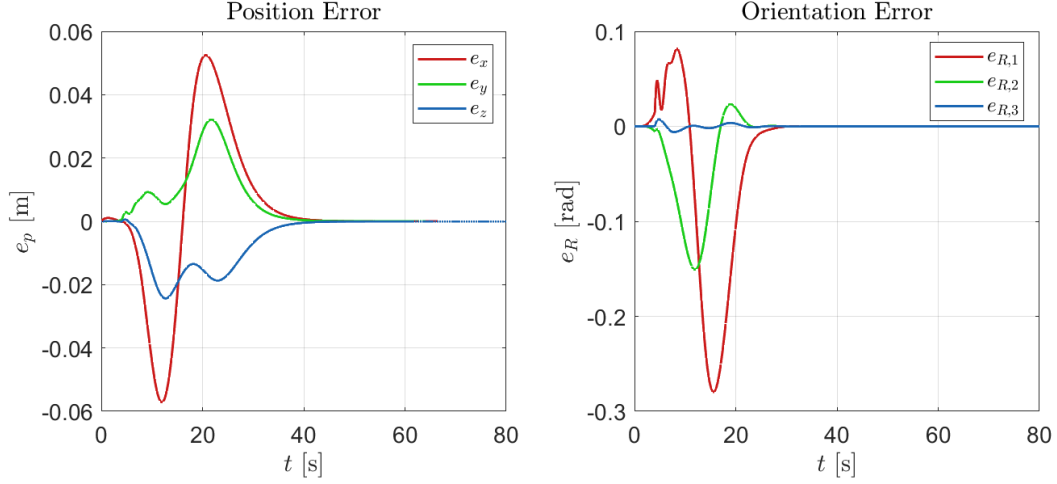


Figure 3.18: Behavior of the OpenAUV with the **current**, 1.5 **Step** force and moment and PID action with the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ .

### 3.c Advanced Nonlinear Thruster Hydrodynamic Modeling

Up to this point, the thruster dynamics have been modeled using a simplified linear approach. Specifically, the required thrust vector was computed by multiplying the control wrench by the pseudoinverse of the static allocation matrix. By employing the Moore-Penrose pseudoinverse, we optimized the control effort by minimizing the total power consumption. Subsequently, the desired angular velocity  $\Omega_{des,i}$  of the  $i$ -th propellers was derived assuming a purely quadratic relationship between the individual thrust  $T_i$  (where  $T_i \equiv u_i$  represents the  $i$ -th component of the control input vector  $\mathbf{u}$ ) and the rotational speed:

$$T_i = \begin{cases} c_{f,forward} \Omega_{des,i} |\Omega_{des,i}| & \text{if } \Omega_{des,i} \geq 0 \\ c_{f,backward} \Omega_{des,i} |\Omega_{des,i}| & \text{if } \Omega_{des,i} < 0 \end{cases} \quad \text{for } i = 1, \dots, 8 \quad (3.7)$$

However, this simplification neglects critical hydrodynamic phenomena that occur when propellers operate in a fluid flow, particularly at high vehicle velocities. To address these limitations, we implemented a more accurate nonlinear model based on the work of [Kim and Chung \(2006\)](#). The primary effect neglected in our previous linear model is the dependency of the generated thrust on the local water inflow velocity,  $v_{a,i}$ . As the vehicle's forward speed increases, the effective thrust produced by the thrusters decreases for a constant angular velocity  $\Omega_{des,i}$ .

#### 3.c.1 Hydrodynamic Degradation and Advance Ratio

The implemented model characterizes thrust not merely as a quadratic function of rotational speed, but as a function of the thrust coefficient  $K_{T,i}$ , defined as:

$$K_T(J_i) = \frac{T_i}{\rho D^4 n_i |n_i|} \quad (3.8)$$

where  $n_i = \Omega_{des,i}/(2\pi)$  for  $i = 1, \dots, 8$  represents the rotational speed in revolutions per second,  $D$  is the propeller diameter, set to 76 mm for the *BlueRobotics T200* thrusters, and  $J_i$  is the

*Advance Ratio* for  $i$ -th propeller, calculated as:

$$J_i = \frac{v_{a,i}}{n_i D} \quad (3.9)$$

The nominal thrust coefficient  $K_{T,i}$  is subject to degradation according to the following relation:

$$K_{T,i} = K_{T0,i} - a_1 J_i - a_2 J_i |J_i| \quad (3.10)$$

where  $a_1 = 0.25$  and  $a_2 = 0.10$  are the nonlinear degradation coefficients quantifying the thrust decay as the vehicle advances. The nominal coefficient  $K_{T0,i}$  is updated in real-time based on the propeller's rotation direction:

$$K_{T0,i} = \begin{cases} \frac{4\pi^2 c_{f,forward}}{\rho D^4} & \text{if } \Omega_{des,i} \geq 0 \\ \frac{4\pi^2 c_{f,backward}}{\rho D^4} & \text{if } \Omega_{des,i} < 0 \end{cases} \quad \text{for } i = 1, \dots, 8 \quad (3.11)$$

### 3.c.2 Wake Fraction and Inflow Velocity

To account for thruster-hull interaction, we incorporate the *wake fraction* coefficient  $w = 0.1$ . This parameter models the fact that the thruster does not operate in static fluid, but rather within the wake created by the vehicle's motion. The local advance velocity  $v_{a,i}$  is computed by projecting the body-frame linear velocity onto the thruster's axis:

- **Horizontal Thrusters** ( $1 \leq i \leq 4$ ):  $v_{a,i} = (v_x^b d_{x,i} + v_y^b d_{y,i}) (1 - w)$  (combined Surge and Sway motion);
- **Vertical Thrusters** ( $5 \leq i \leq 8$ ):  $v_{a,i} = v_z^b (1 - w)$  (Heave motion).

As the vehicle velocity increases,  $v_{a,i}$  increases, leading to a higher  $J_i$  value. Consequently, the negative terms in the  $K_{T,i}$  equation reduce the overall thrust coefficient. To ensure numerical stability and physical consistency, a lower bound of 0.01 is applied to  $K_{T,i}$ . The final effective thrust  $T_{real,i}$  is thus computed as:

$$T_{real,i} = \rho \cdot n_i |n_i| \cdot D^4 \cdot K_{T,i} \quad (3.12)$$

In the following simulations, two distinct trajectories were considered: a **slow trajectory** and a **fast trajectory**. In the slow scenario, the AUV travels from the initial position ( $x_0 = 0, y_0 = 0, z_0 = -1$ ) to the final target ( $x_f = 1, y_f = 1, z_f = -4$ ) over a total duration of 22 seconds. Conversely, for the fast trajectory, the vehicle must navigate from ( $x_0 = 0, y_0 = 0, z_0 = -1$ ) to ( $x_f = 5, y_f = 5, z_f = -6$ ) within 8 seconds. Consequently, the AUV reaches significantly higher velocities before the final hovering phase, as it must cover a longer distance in a shorter timeframe. The impact of the system's non-linearities is particularly evident in the fast trajectory, where the inflow velocity effect becomes the dominant factor in thruster performance degradation.

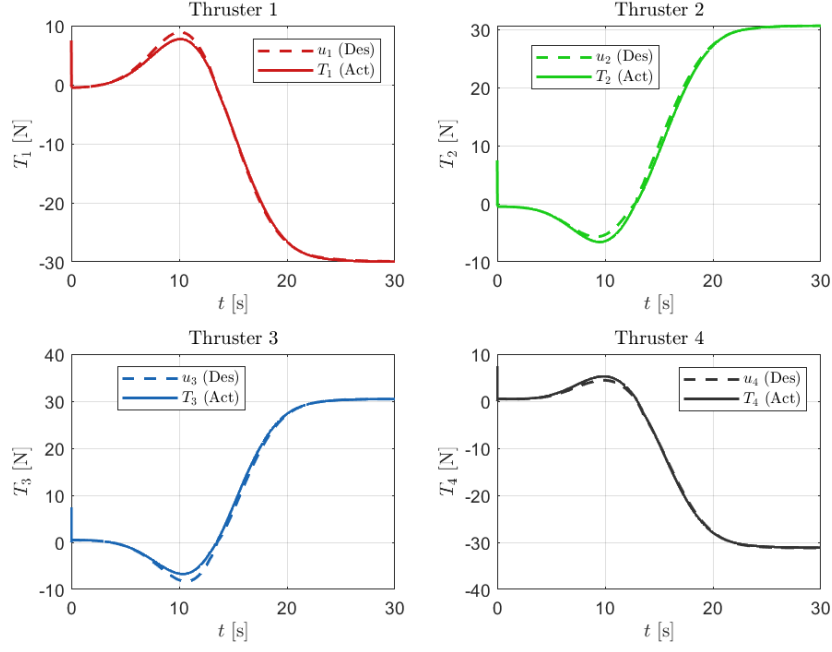


Figure 3.19: Time history of  $T_{\text{real},i}$  and  $u_i$  for  $i = 1, \dots, 4$  of the OpenAUV with the **current**, PID action and the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ . Slow trajectory case.

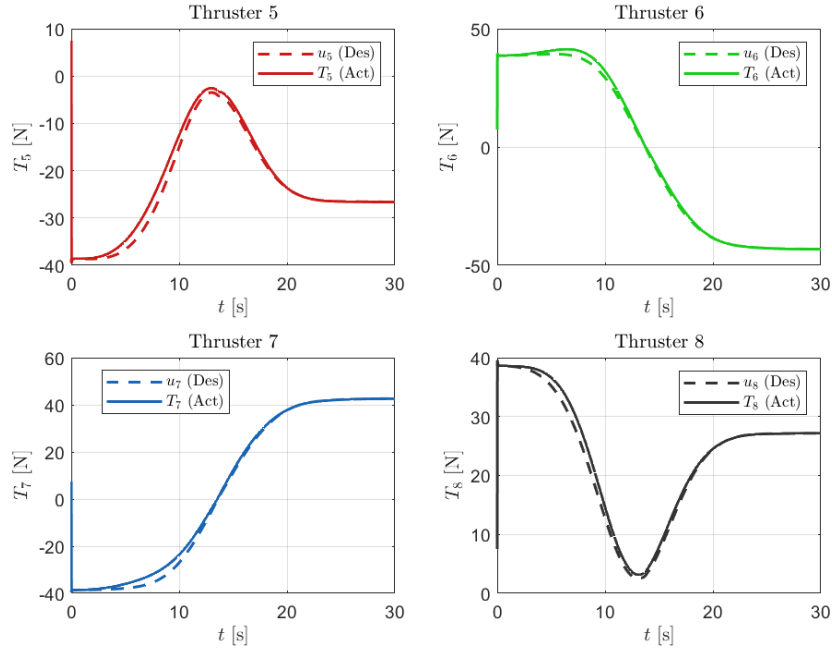


Figure 3.20: Time history of  $T_{\text{real},i}$  and  $u_i$  for  $i = 5, \dots, 8$  of the OpenAUV with the **current**, PID action and the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ . Slow trajectory case.

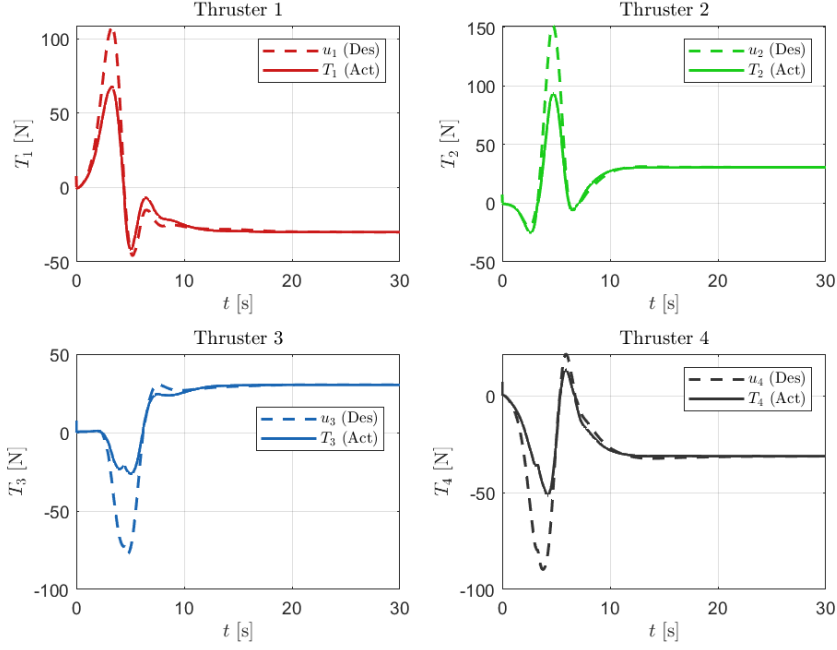


Figure 3.21: Time history of  $T_{\text{real},i}$  and  $u_i$  for  $i = 1, \dots, 4$  of the OpenAUV with the **current**, PID action and the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ . Fast trajectory case.

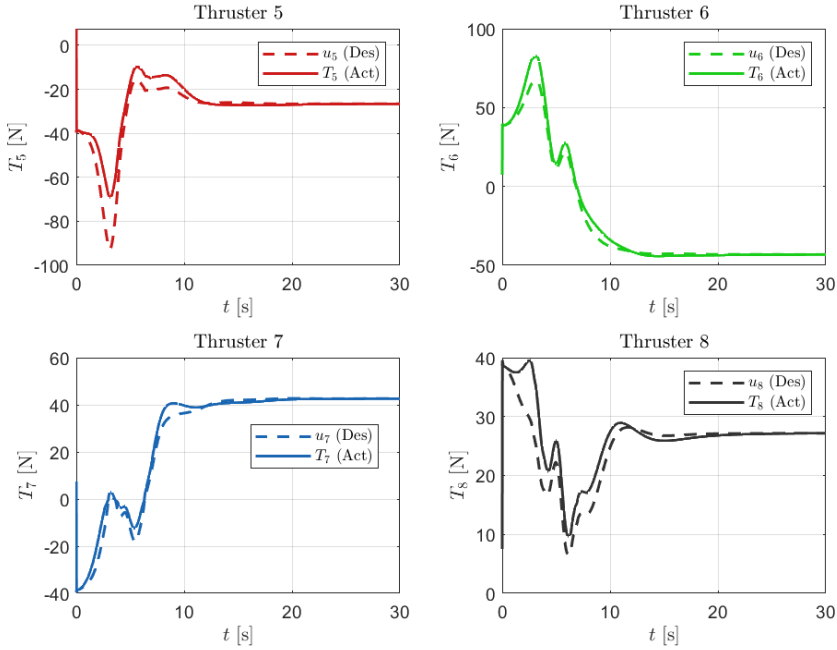


Figure 3.22: Time history of  $T_{\text{real},i}$  and  $u_i$  for  $i = 5, \dots, 8$  of the OpenAUV with the **current**, PID action and the Momentum-based Estimator,  $\mathbf{K}_i = 0.1\mathbf{I}_3$  and  $\mathbf{K}_{iR} = \text{diag}(3, 1, 1)$ . Fast trajectory case.

The simulations were obtained using the Simulink file `OpenAUV_v20BS.slx` as a reference.





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- Sha, Z., Wang, X., Yang, M., Lei, H., and Zhang, F. (2025). A portable autonomous underwater vehicle with multi-thruster propulsion: Design, development, and vision-based tracking control. *IEEE Robotics and Automation Letters*, 10(4):3046–3053.